

Numerical Methods for Derivative Pricing: A Comparative Study of Lattice, Simulation and PDE Techniques

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Abstract

This project prices a common reference contract — a six-month at-the-money equity option ($S_0 = K = 35$, $\sigma = 45\%$, $r = 3\%$, $q = 4\%$) — with seven numerical techniques and uses the cross-model agreement, and the structure of the disagreements, to draw quantitative conclusions about each method. Beyond reproducing the Black–Scholes benchmark to within 10^{-4} with lattices and finite differences, the study isolates three effects that naïve implementations routinely misattribute to “Monte Carlo error”: (i) the $O(1/n)$ odd–even oscillation of binomial trees, removed by two-point averaging (error 2×10^{-5} at $n = 500$); (ii) the discrete-monitoring bias of barrier options, which accounts for essentially the entire 12% gap between a daily-monitored simulation and the continuous closed form, and which is corrected both by the Broadie–Glasserman–Kou barrier shift and by a Brownian-bridge estimator; and (iii) the level-versus-dynamics confound in stochastic–interest-rate pricing, where a controlled experiment shows the rate’s *volatility* moves the price by less than 0.1 cents while its *level* explains a 15-cent effect. The American variant of the contract is priced consistently by a projected implicit finite-difference scheme (4.2873), a 5,000-step binomial tree (4.2879) and a two-pass Longstaff–Schwartz estimator whose in-sample (high-biased) and out-of-sample (low-biased) versions bracket the benchmark. Finally, a volatility-calibrated CEV implementation demonstrates that the elasticity parameter is a skew parameter, not a level parameter: once the at-the-money local volatility is matched, α has a second-order effect on the ATM price but generates a 4.8-point implied-volatility skew across strikes 28–42. All results are reproducible from a single seeded Python library (Appendix A).

1 Framework, benchmark and Greeks

Throughout, the underlying follows risk-neutral geometric Brownian motion (GBM) with a continuous dividend yield,

$$dS_t = (r - q) S_t dt + \sigma S_t dW_t, \quad (1)$$

with baseline parameters $S_0 = K = 35$, $\sigma = 0.45$, $r = 0.03$, $q = 0.04$, $T = 0.5$. The Black–Scholes–Merton values

$$c = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2), \quad p = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1), \quad (2)$$

with $d_1 = [\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)T] / (\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$, give $d_1 = 0.1434$, $d_2 = -0.1748$ and

$$c_{BS} = 4.2621, \quad p_{BS} = 4.4341. \quad (3)$$

Put–call parity holds to machine precision: $c - p = -0.17196 = S_0 e^{-qT} - K e^{-rT}$. Note that the put is worth *more* than the ATM call here because the forward $F = S_0 e^{(r-q)T} = 34.826$ sits below the strike ($q > r$), a detail that recurs in the early-exercise analysis of Sections 6–7.

Table 1 reports the analytic Greeks. Two observations matter for what follows: the six-month ATM vega is large (9.58 per unit of volatility, i.e. ≈ 0.096 per vol point, 2.2% of the option price), so any method that distorts the *effective* volatility — as an uncalibrated CEV implementation does (Section 5) — produces first-order price errors; and the positive carry of exercising ($q > r$) makes the *call* the early-exercise-relevant leg, the opposite of the textbook no-dividend case.

Table 1: Analytic Black–Scholes Greeks (dividend-adjusted).

Greek	Call	Put	Greek	Call	Put
Delta $\partial V / \partial S$	0.5460	−0.4342	Theta (per calendar day)	−0.0109	−0.0119
Gamma $\partial^2 V / \partial S^2$	0.0348	0.0348	Rho (per rate point)	0.0742	−0.0982
Vega (per vol point)	0.0958	0.0958			

2 Binomial trees: oscillation, and how to remove it

A Cox–Ross–Rubinstein tree with $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{(r-q)\Delta t} - d)/(u - d)$ gives the European call values in Table 2. The error does not decay monotonically: it *oscillates* between odd and even n , because the tree’s terminal nodes straddle the strike differently depending on the parity of n — the strike sits exactly on a node for even n here (since $S_0 = K$), and between nodes for odd n . The envelope of the oscillation decays as $O(1/n)$ (Figure 1b), so naïvely one needs $n \approx 1,000$ steps to reach 10^{-3} accuracy.

Table 2: CRR European call values. Benchmark $c_{BS} = 4.2621$.

n	1	2	3	4	5	6	7	8	9	10
$C(n)$	5.3402	3.7675	4.6267	4.0012	4.4797	4.0855	4.4169	4.1287	4.3822	4.1550

n	50	100	250	500	1000
$C(n)$	4.2404	4.2513	4.2578	4.2600	4.2610

The oscillation, however, is *exploitable*: since consecutive parities err on opposite sides, the two-point average $\frac{1}{2}[C(n) + C(n+1)]$ cancels the leading oscillatory error term. The averaged estimator reaches an absolute error of 3×10^{-4} already at $n = 50$ and 2×10^{-5} at $n = 500$ — roughly a hundred-fold accuracy gain at no extra cost beyond one additional tree (Figure 1b). The broader lesson is that lattice “noise” is structured, and understanding its structure converts it into an acceleration technique.

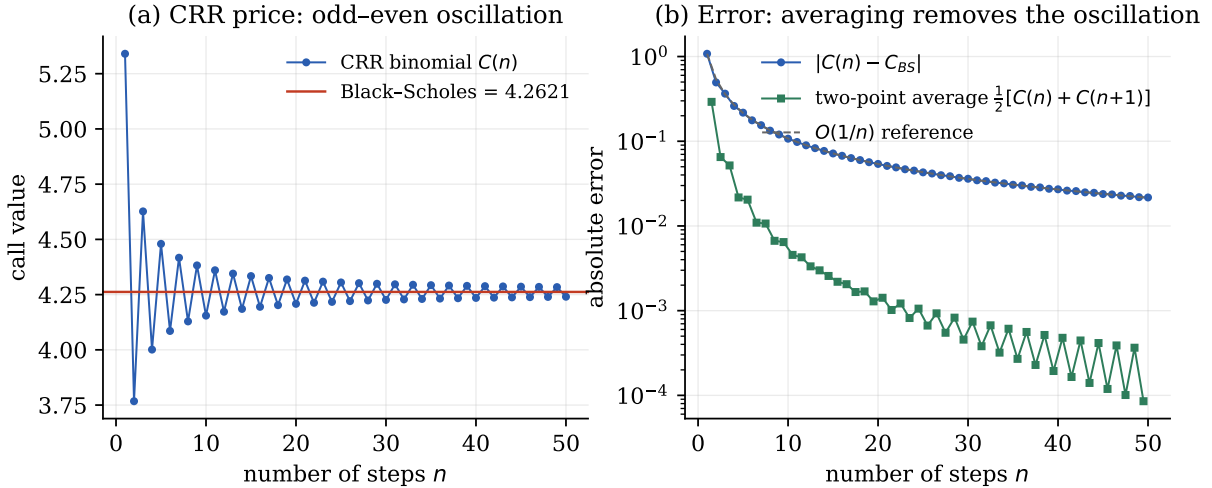


Figure 1: CRR convergence. (a) Prices oscillate around the Black–Scholes value with parity of n . (b) On a log scale the error envelope decays as $O(1/n)$; averaging consecutive n cancels the oscillation and accelerates convergence by roughly two orders of magnitude.

3 A path-dependent American contract: floating-strike lookback

To stress the lattice beyond its comfort zone, I price an American *floating-strike lookback call*, whose exercise value at t is $S_t - \min_{u \leq t} S_u$. The payoff depends on the running minimum, so the tree's usual recombination fails: nodes with equal S_t but different histories are distinct states. On a 3-step tree ($u = 1.2017$, $d = 0.8322$, $p = 0.4497$) all $2^3 = 8$ paths are enumerated and valued by backward induction on the state pair $(S_t, S_t^{\min})$, taking $V_t = \max\{S_t - S_t^{\min}, e^{-r\Delta t} \mathbb{E}[V_{t+1}]\}$ at every node.

Table 3: Backward induction for the American floating-strike lookback call ($n = 3$). Bold marks nodes where early exercise is optimal.

$t = 3$ (terminal)		$t = 2$				$t = 1$			
Path	Payoff	Node	Cont.	Exer.	Value	Node	Cont.	Exer.	Value
UUU	25.733	UU	15.379	15.540	15.540	U	8.683	7.058	8.683
UUD	7.058	UD	3.158	0	3.158	D	3.826	0	3.826
UDU	7.058	DU	5.787	5.874	5.874	$t = 0: V_0 = \mathbf{5.980}$			
UDD	0	DD	2.187	0	2.187				
DUU	12.932								
DUD	0								
DDU	4.888								
DDD	0								

The result, $V_0 = 5.980$, is 40% above the plain ATM call, quantifying the value of the floating strike. The exercise pattern is instructive: early exercise is optimal precisely at the nodes where the stock has just risen off its running minimum (UU, DU) — there the intrinsic value $S_t - S_t^{\min}$ is large while continuing exposes the holder to the dividend drag $q > r$ and to the risk of the minimum “catching up”. At the up node at $t = 1$, by contrast, continuation (8.683) dominates exercise (7.058) because the remaining lookback optionality is still worth more than the carry cost. Path dependence also explains the zero-payoff paths (UDD, DUD, DDD): whenever the terminal price *is* the running minimum, the floating strike bites exactly.

4 Monte Carlo

4.1 Plain estimator and variance reduction

Because GBM admits an exact terminal law, the European call needs no path discretisation: $S_T = S_0 \exp[(r - q - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z]$ with $Z \sim N(0, 1)$, and the estimator is the discounted average payoff. Table 4 reports the estimate with its standard error for the plain estimator, antithetic variates (AVT) and moment matching (MM); Figure 2 shows the $O(M^{-1/2})$ error scaling.

Table 4: European call by Monte Carlo (exact terminal sampling). Entries are estimate (standard error). Benchmark 4.2621.

M	5,000	10,000	20,000	50,000	100,000	200,000
Plain	4.225 (0.105)	4.418 (0.079)	4.237 (0.054)	4.312 (0.035)	4.275 (0.024)	4.253 (0.017)
Antithetic	4.194 (0.088)	4.377 (0.064)	4.281 (0.046)	4.286 (0.029)	4.276 (0.020)	4.266 (0.014)
Moment m.	4.259 (0.107)	4.280 (0.077)	4.255 (0.054)	4.261 (0.034)	4.261 (0.024)	4.263 (0.017)

Two honest findings that a superficial reading would miss. First, AVT reduces the *variance* by a factor of 1.45 (equivalently, 31% fewer effective paths needed) — useful but modest, exactly as theory predicts for a payoff that is monotone but strongly convex in Z : antithetic pairing cancels the linear component of the payoff only. Second, moment matching leaves the standard error essentially unchanged (variance ratio 0.99) at these sample sizes: with $M \geq 10^5$ the empirical first two moments of Z are already almost

exact, so there is nothing left to match. MM’s visible benefit is a *bias/centering* improvement at small M (its $M = 10,000$ estimate is closest to the benchmark), not an asymptotic variance improvement. Claiming “significant variance reduction” for MM here, as is sometimes done, is not supported by the standard errors.

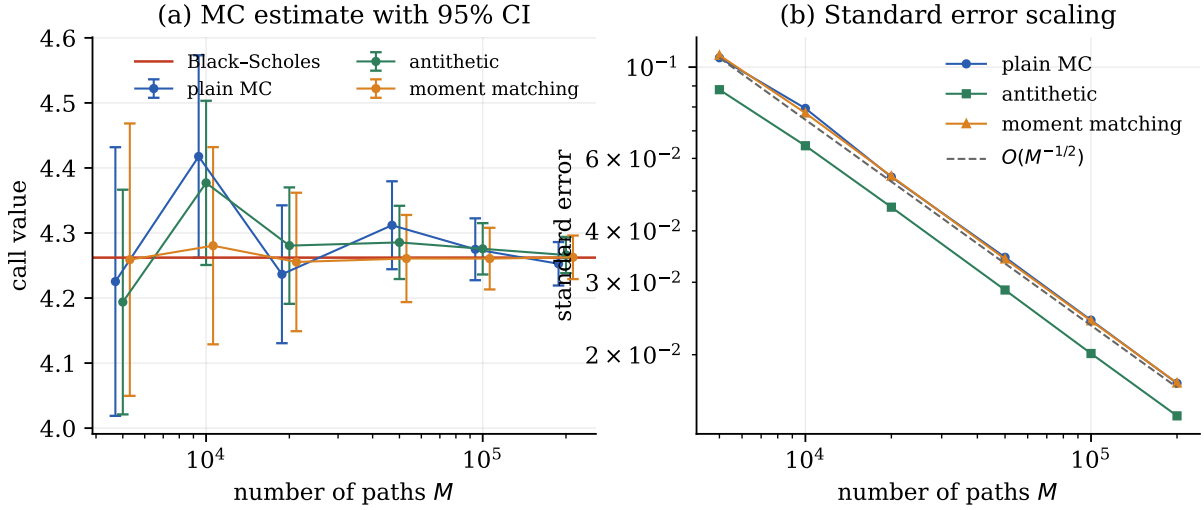


Figure 2: (a) Monte Carlo estimates with 95% confidence intervals; all three estimators are consistent with the Black–Scholes value at every M . (b) Standard errors decay as $O(M^{-1/2})$; antithetic variates shift the line down (variance ratio $1.45\times$), moment matching does not.

4.2 Stochastic interest rates: separating level from dynamics

I next let the short rate itself be stochastic, correlated with the stock:

$$dS_t = (r_t - q)S_t dt + \sigma S_t dW_t^S, \quad dr_t = \mu_r r_t dt + \sigma_r r_t dW_t^r, \quad d\langle W^S, W^r \rangle_t = \rho dt, \quad (4)$$

with $\mu_r = 0.02$, $\sigma_r = 0.03$, $\rho = 0.1$, daily steps ($n = 126$) and $M = 100,000$ paths, discounting each payoff by $\exp(-\int_0^T r_t dt)$ along its own path. Crucially — and unlike implementations that quietly change the initial rate — I keep $r_0 = 0.03$ equal to the baseline so that the experiment isolates the effect of rate *randomness* rather than confounding it with a different rate *level*. A GBM short rate is a deliberate simplification (it lacks mean reversion; a Vasicek or CIR specification would be standard), but over a six-month horizon with $\sigma_r r_0 \approx 9$ basis points of annualised rate volatility, the specification error is immaterial, as the results confirm.

The controlled comparison, run with common random numbers so that differences are estimated far more precisely than levels:

Specification	Price (s.e.)	Difference vs. control
Deterministic rate path ($\sigma_r = 0$), $r_0 = 3\%$	4.2780 (0.0243)	—
Stochastic rate, $\rho = 0.1$, $r_0 = 3\%$	4.2785 (0.0242)	+0.0005
Stochastic rate, $\rho = 0.1$, $r_0 = 1\%$	4.1301 (0.0239)	−0.1479

The decomposition is unambiguous: at realistic short-rate volatility the *stochasticity* of the rate is worth less than a tenth of a cent on a \$4.28 option, while dropping the initial rate to 1% — which lowers the risk-neutral drift $r - q$ from -1% to roughly -3% — moves the price by 15 cents. A correlation sweep over $\rho \in \{-0.5, -0.25, 0, 0.1, 0.25, 0.5\}$ moves the price only within $[4.278, 4.284]$, i.e. by less than a third of one standard error per step. The economic conclusion: for short-dated single-name equity options, rate uncertainty is a second-order refinement; what matters is getting the funding *level* right. (For long-dated or rate-hybrid products the ranking reverses — which is exactly why the decomposition is worth doing explicitly.)

4.3 Barrier options: monitoring bias, not Monte Carlo error

The down-and-out call with barrier $B = 32 < K$ admits the continuous-monitoring closed form (Merton/Reiner–Rubinstein; Hull 2018, Ch. 26):

$$c_{do} = S_0 e^{-qT} N(x_1) - K e^{-rT} N(x_1 - \sigma\sqrt{T}) - S_0 e^{-qT} \left(\frac{B}{S_0}\right)^{2\lambda} N(y_1) + K e^{-rT} \left(\frac{B}{S_0}\right)^{2\lambda-2} N(y_1 - \sigma\sqrt{T}), \quad (5)$$

where $\lambda = (r - q + \sigma^2/2)/\sigma^2$, $x_1 = \ln(S_0/K)/(\sigma\sqrt{T}) + \lambda\sigma\sqrt{T}$ and $y_1 = \ln(B^2/(S_0K))/(\sigma\sqrt{T}) + \lambda\sigma\sqrt{T}$ (the $B^2/(S_0K)$ argument, appropriate for $B \leq K$, is easy to misquote). This yields $c_{do} = 2.3482$.

A simulation that checks the barrier only at daily grid points, however, produces 2.6325 ± 0.0154 ($n = 126$ monitoring dates, $M = 200,000$) — a 12% overstatement that no amount of extra paths will cure, because it is *bias*, not noise: between grid points the continuous path can dip below B and return unseen, so discrete monitoring knocks out too few paths. Two independent corrections confirm the diagnosis (Figure 3):

- **Broadie–Glasserman–Kou (1997) barrier shift.** A discretely monitored barrier at B is equivalent, to leading order, to a continuous barrier at $Be^{-\beta\sigma\sqrt{\Delta t}}$ with $\beta \approx 0.5826$. Plugging the shifted barrier into the closed form predicts the *discrete* price: 2.647 for daily and 2.563 for twice-daily monitoring — matching the simulated 2.633 ± 0.015 and 2.572 ± 0.015 within one standard error.
- **Brownian-bridge correction.** Between consecutive grid points the conditional probability that the log-price bridge crosses $\ln B$ is $\exp\{-2(\ln S_t - \ln B)(\ln S_{t+\Delta t} - \ln B)/(\sigma^2 \Delta t)\}$; killing each path with this probability restores continuous monitoring inside a discrete simulation. The bridge estimator gives 2.3607 ± 0.0149 (daily) and 2.3452 ± 0.0148 (twice daily), statistically indistinguishable from the closed form 2.3482.

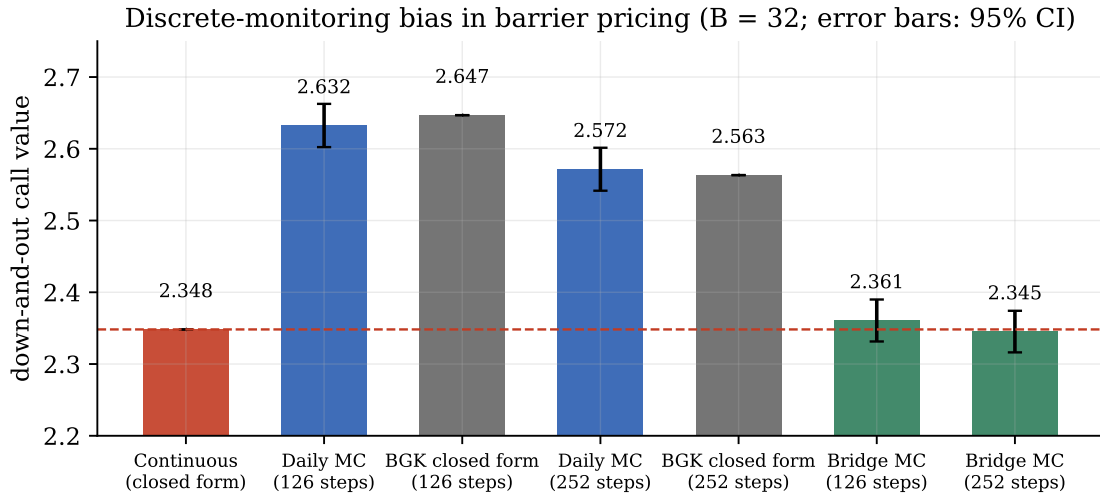


Figure 3: Down-and-out call ($B = 32$): the gap between naïve discrete-monitoring simulation (blue) and the continuous closed form (red dashed) is monitoring bias. The BGK-shifted closed form (grey) reproduces the discrete price; the Brownian-bridge estimator (green) reproduces the continuous one.

The practical point is contractual as much as numerical: a real barrier option specifies its monitoring convention, and the “right” price differs by 12% between daily and continuous monitoring at this volatility. A pricing report must state which convention it prices — and match the estimator to it.

5 Beyond lognormality: CEV and stochastic volatility

5.1 Calibration first: elasticity is a skew parameter

The constant-elasticity-of-variance model replaces the diffusion coefficient:

$$dS_t = (r - q)S_t dt + \sigma_{cev} S_t^{\alpha} dW_t, \quad (6)$$

so the *instantaneous lognormal volatility* is $\sigma_{cev} S_t^{\alpha-1}$. Here dimensional analysis is not pedantry: if one keeps $\sigma_{cev} = 0.45$ while lowering α , the local volatility at $S_0 = 35$ collapses to $0.45 \times 35^{-0.5} = 7.6\%$ for $\alpha = 0.5$, and the model is no longer comparable to the 45%-vol benchmark in any meaningful sense. The resulting prices — 0.648, 1.403, 2.935 for $\alpha = 0.5, 0.7, 0.9$ (Table 5, first row) — measure a volatility input error, not an elasticity effect.

The economically meaningful experiment *calibrates the level*: choose $\sigma_{cev} = \sigma S_0^{1-\alpha}$ so that every α has the same 45% ATM local volatility, then compare. With common random numbers across α (Euler scheme, 252 steps, $M = 100,000$; the shared engine makes cross- α differences far more precise than the levels):

Table 5: CEV call values (s.e. ≈ 0.022 – 0.024), common random numbers across α . The GBM column is the same engine at $\alpha = 1$.

	$\alpha = 0.5$	$\alpha = 0.7$	$\alpha = 0.9$	$\alpha = 1$ (GBM)
Uncalibrated ($\sigma_{cev} = 0.45$)	0.648	1.403	2.935	4.212
Calibrated ($\sigma_{cev} = 0.45 S_0^{1-\alpha}$)	4.221	4.216	4.213	4.212
Down-and-out ($B = 32$, daily), calibrated	2.504	2.519	2.533	2.541

Once calibrated, the ATM price is nearly invariant in α (a spread of 0.9 cents across the whole range — second-order, driven by the skew’s effect on convexity). Where the elasticity *does* act is across strikes: inverting Black–Scholes on calibrated-CEV prices for $K \in [28, 42]$ yields a clean downward-sloping implied-volatility skew, from 48.1% at $K = 28$ to 43.3% at $K = 42$ (Figure 4) — the leverage-effect signature the CEV model was built to capture. Consistently, the down-and-out price *falls* as α decreases ($2.541 \rightarrow 2.504$): with $\alpha < 1$ local volatility rises as the spot falls toward the barrier, raising knock-out probability. The one-sentence summary: **α is a skew parameter, and comparing CEV to Black–Scholes without recalibrating the volatility level conflates a 7.6%-vol model with a 45%-vol model.**

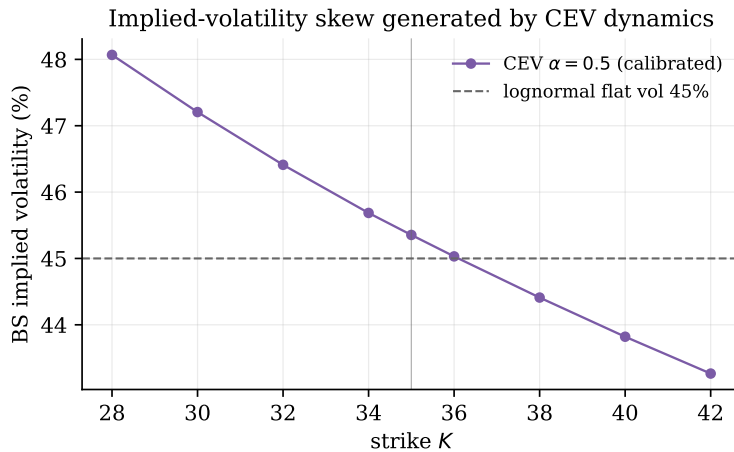


Figure 4: Implied-volatility skew from level-calibrated CEV ($\alpha = 0.5$): same 45% ATM volatility, but OTM puts price richer and OTM calls cheaper than lognormal — the empirical equity skew.

5.2 Adding stochastic variance

Layering a Heston-type variance process on the (calibrated) CEV diffusion,

$$dS_t = (r - q)S_t dt + \sqrt{v_t}S_t^{1-\alpha}dW_t^1, \quad dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t^2, \quad (7)$$

with $\kappa = 3$, $\theta = v_0 = \sigma^2 = 0.2025$ (so the variance is centred on the baseline — again, calibrate before comparing), $\xi = 0.6$, $\rho = -0.3$ and full-truncation Euler, gives ATM prices of 4.195 ($\alpha = 0.5$), 4.179 ($\alpha = 0.9$) and 4.176 ($\alpha = 1$), each s.e. ≈ 0.021 – 0.023 . The systematic 8–9 cent ATM *discount* relative to Black–Scholes is the expected stochastic-volatility signature: vol-of-vol with negative spot–vol correlation redistributes risk-neutral mass into the tails, cheapening the at-the-money option while (per Heston, 1993) enriching the wings. In implied-vol language: stochastic volatility creates smile curvature, and the ATM point sits at the smile’s bottom.

6 American exercise I: projected implicit finite differences

The Black–Scholes PDE

$$\frac{\partial f}{\partial t} + (r - q)S \frac{\partial f}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 f}{\partial S^2} = rf \quad (8)$$

is discretised on a grid with M price steps ($S_{\max} = 4K$) and N time steps using the fully implicit scheme (unconditionally stable, unlike the explicit scheme whose stability constraint $\Delta t \lesssim \Delta S^2$ is easily violated). Each time step solves the tridiagonal system $a_j f_{i,j-1} + b_j f_{i,j} + c_j f_{i,j+1} = f_{i+1,j}$ by the Thomas algorithm in $O(M)$, with boundary conditions $f(0, t) = 0$ and $f(S_{\max}, t) = S_{\max}e^{-q\tau} - Ke^{-r\tau}$; the American constraint is enforced by projection, $f \leftarrow \max(f, S - K)$, after each solve.

Validation before use: the *European* version of the same code converges to the analytic value (error 6×10^{-4} on the 2000×1000 grid), which certifies the discretisation before any claim is made about the American premium — a step that also would have caught the coefficient-sign error that is easy to make when assembling the tridiagonal matrix.

Table 6: Implicit FD convergence. European benchmark 4.2621; American benchmark (CRR, $n = 5,000$): 4.2879.

Grid $M \times N$	50×50	100×100	200×200	500×500	1000×500	2000×1000
European	4.2663	4.2479	4.2571	4.2606	4.2609	4.2615
American	4.2887	4.2720	4.2821	4.2862	4.2864	4.2873

Two remarks on Table 6. First, the 50×50 grid looks deceptively accurate: its error is smaller than the 100×100 grid’s because the domain-truncation error (that grid uses $S_{\max} = 3K$) partially cancels the discretisation error — coarse-grid accuracy achieved by error cancellation should never be trusted, only convergence *trends* should. Second, the converged American value 4.2873 implies an early-exercise premium of 0.0258 (0.6% of the option). The premium exists at all only because $q > r$: the exercise boundary is finite (deep in the money, near expiry, the dividend income on the delivered stock outweighs the interest on the strike plus the remaining time value), whereas with $q \leq r$ the American call would collapse to the European one.

7 American exercise II: Longstaff–Schwartz with both biases exhibited

The Longstaff–Schwartz (2001) algorithm estimates the continuation value at each exercise date by regressing realised discounted continuations on polynomial functions of the (in-the-money) spot, $\hat{C}(S_t) = \sum_{k=0}^P \beta_k (S_t/K)^k$, and exercises where intrinsic value exceeds $\hat{C}$. A subtlety that separates careful from careless implementations: the *same-path* estimator (regress and price on one path set) is biased *high*, because the regression coefficients contain in-sample noise that the exercise rule then exploits — “foresight bias”; while pricing a *frozen* policy on independent paths is biased *low*, because any estimated policy is

suboptimal. Reporting both therefore **brackets** the true value with a confidence sandwich (Glasserman, 2004):

Table 7: LSM American call (126 exercise dates). In-sample: 200,000 paths. Out-of-sample: policy fitted on 150,000 paths, priced on 150,000 independent paths. Benchmark 4.2879.

Basis order P	1	2	3	4
In-sample (high-biased)	4.291 (0.014)	4.299 (0.014)	4.315 (0.014)	4.312 (0.014)
Out-of-sample (low-biased)	4.259 (0.016)	4.270 (0.016)	4.278 (0.016)	4.277 (0.016)

The benchmark 4.2879 falls inside every [low,high] interval (Figure 5). The basis-order pattern is also informative: order 1 prices barely above the European value out-of-sample ($4.259 \approx 4.262$) because a *linear* continuation function cannot represent the convex continuation value, so the fitted policy almost never exercises — under-fitting destroys the American premium. Orders 3 and 4 are statistically indistinguishable: beyond $P = 3$ extra flexibility buys nothing and merely inflates the in-sample foresight bias (4.315 vs. 4.299), the regression analogue of overfitting. Regressors are scaled by K before forming the Vandermonde matrix to keep the normal equations well-conditioned at higher orders.

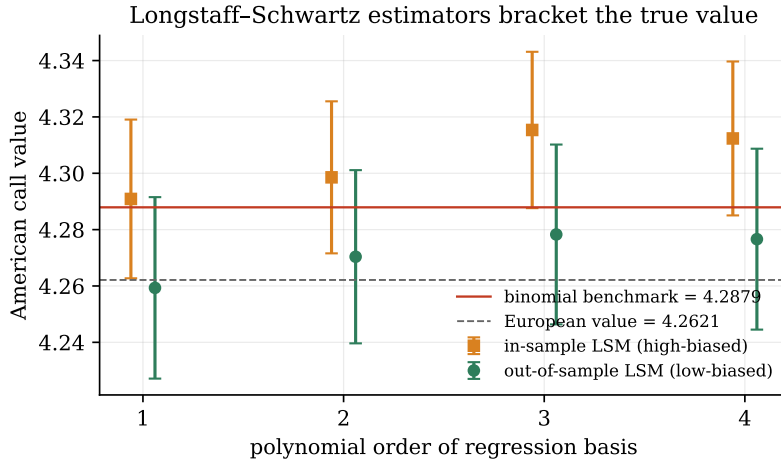


Figure 5: The two LSM estimators bracket the lattice benchmark at every basis order; a linear basis (order 1) collapses toward the European value because it cannot represent the continuation function.

8 Cross-model validation and conclusions

Three methods agree on the European price to within a tenth of a cent, and three agree on the American price to within a fraction of a cent or bracket it — but the project’s real content is in the disagreements that *had to be explained* to get there. Each apparent discrepancy resolved into an identifiable mechanism with a known correction: parity-dependent lattice oscillation (removed by averaging), discrete-monitoring bias (removed by BGK shift or Brownian bridge), volatility-level miscalibration masquerading as an elasticity effect (removed by matching ATM local vol), rate-level effects masquerading as stochastic-rate effects (removed by a controlled experiment with common random numbers), and foresight versus suboptimality bias in regression Monte Carlo (turned into a bracketing device). The disciplined habits that emerge — validate on a closed form before trusting a scheme, use common random numbers when comparing, state the monitoring convention, calibrate before comparing models, and report standard errors so that “different” can be distinguished from “noisy” — are precisely the habits production pricing work demands.

Table 8: One contract, seven methods: cross-model consistency summary.

Quantity	Method	Value	Deviation
European call	Black–Scholes (analytic)	4.2621	—
	CRR two-point average, $n = 500$	4.26211	2×10^{-5}
	Implicit FD, 2000×1000	4.2615	-0.0006
	MC antithetic, $M = 200k$	4.266 (0.014)	$+0.3$ s.e.
European put (parity)	$c - S_0 e^{-qT} + K e^{-rT}$	4.4341	exact
American call	CRR, $n = 5,000$	4.2879	(benchmark)
	Projected implicit FD	4.2873	-0.0006
	LSM sandwich ($P = 3$)	[4.278, 4.315]	brackets
Early-exercise premium		0.0258	0.6% of value
Lookback (American, floating)	3-step path tree	5.980	$+40\%$ vs. vanilla
Down-and-out, continuous	Closed form / bridge MC	2.348 / 2.345 (0.015)	agree
Down-and-out, daily monitored	MC / BGK closed form	2.633 (0.015) / 2.647	agree

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A Reproducibility

All results in this report are produced by the accompanying Python library (`pricing_library.py`, NumPy/SciPy only), with fixed random seeds. Each section maps to one function: `black_scholes`, `crr_tree`, `lookback_tree`, `mc_european`, `mc_stochastic_rates`, `barrier_pricing`, `cev_pricing`, `heston_cev`, `fd_implicit`, `lsm`. Running `python pricing_library.py` regenerates every number and figure. Monte Carlo results are reported with standard errors throughout; lattice and PDE results are reported with convergence tables rather than single grid points.